

battery is used as the power supply.

The CD4047 (IC1) is wired in astable multivibrator mode, requiring only a resistor and a capacitor to produce a square wave with a 50% duty cycle. The frequency is determined by the resistor and capacitor connected across pins 1, 2, and 3. The signals at pins 10 and 11 are inverted relative to each other. The output frequency is set by resistor R1 and capacitor C1. Here, R1 is 100 kilo-ohm, and C1 is 1 microfarad, resulting in a frequency of approximately 2Hz to 3Hz.

The clock output (pin 10) of the CD4047 is connected to the clock input (pin 14) of the 4017 IC. LED1 blinks to confirm that the CD4047 is clocking properly. One end of the nine wires to be tested is connected to test points TP1 through TP9, while their other end is connected to test points TP1-1 through TP9-9. Each wire is paired with an LED, which indicates the wire's condition. If all nine wires are intact, all nine LEDs blink continuously. If a wire is broken, the corresponding LED does not blink, allowing easy identification of the faulty wire.

During testing in EFY lab, nine male-to-male jumpers were used. To check continuity, all nine jumpers were connected to their respective test terminals. When all jumpers were intact, all nine LEDs blinked. If any jumper wire was disconnected (or broken), the corresponding LED did not blink, indicating a fault.

The continuity tester operates using the CD4047, a piezo-buzzer, a BC548 transistor, and two probe terminals (TP10 and TP10-10).

To use the tester, solder red and black probes to terminals TP10 and TP10-10. These probes act as the circuit's input terminals.

PARTS LIST

Semiconductors:

IC1	- CD4047 counter
IC2	- 4017 decade counter
T1	- BC548 NPN transistor
LED1-LED10	- 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 100k Ω
R2-R10, R13	- 470 Ω
R11, R12	- 1k Ω

Capacitors:

C1	- 1 μ F, 25V electrolytic
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Miscellaneous:

PZ1	- Piezo buzzer
S1	- On/off switch
Batt.1	- 9V battery
	- Two probes: red and black
	- 12V battery

When the probes touch each other through a wire, a conductive path, or a PCB track, they send a signal to the base of transistor T1, completing the circuit.

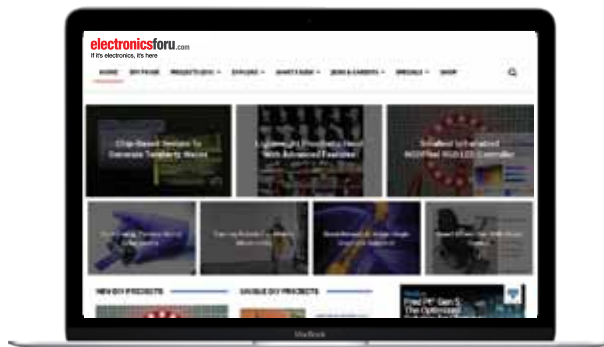
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